

Marketing Analytics Final Project

A: Comprehensive Marketing Analytics Project

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1. Customer Segmentation

Objective:

The objective of this task was to segment customers based on artificial data generated using R and perform k-means clustering to identify distinct customer segments. The segments were then analysed to provide insights and suggest targeted marketing strategies.

Steps Taken:

1. **Data Generation:** We simulated a dataset of 1000 customers with various attributes, including Age, Income, Purchase Frequency, Average Spend, and Preferred Genre. The data was generated using R and the dplyr and tidyr libraries.

The dataset was designed as follows:

- **Age:** Randomly sampled between 18 and 65 years.
- **Income:** Randomly sampled between 20,000 and 100,000.
- **Purchase Frequency:** Randomly sampled between 1 and 20 purchases.
- **Average Spend:** Randomly sampled between 10 and 50.

- **Preferred Genre:** Randomly selected from five genres: Fiction, Non-Fiction, Mystery, Romance, and Sci-Fi.

A new variable, **TotalSpend**, was calculated as $\text{PurchaseFrequency} * \text{AvgSpend}$ to better understand customer spending behaviour.

Output:

```
> # Calculate TotalSpend as PurchaseFrequency * AvgSpend
> customer_data <- customer_data %>%
+   mutate(TotalSpend = PurchaseFrequency * AvgSpend)
>
> # View the first few rows of the data
> head(customer_data)
```

	CustomerID	Age	Income	PurchaseFrequency	AvgSpend	PreferredGenre	TotalSpend
1	1	48	35523	8	29	Mystery	232
2	2	32	27052	17	10	Fiction	170
3	3	31	22221	5	13	Romance	65
4	4	20	40994	9	31	Sci-Fi	279
5	5	59	60458	10	13	Mystery	130
6	6	60	31165	13	31	Fiction	403

2. **K-Means Clustering:** We performed k-means clustering on four key variables: Age, Income, Purchase Frequency, and Average Spend. Based on the results, the dataset was divided into three distinct clusters.

The clustering algorithm was implemented using the `kmeans()` function in R. After running the algorithm, each customer was assigned to one of the three clusters, which were then added as a new column in the dataset.

Output:

```
> kmeans_result <- kmeans(customer_data[, c("Age", "Income", "PurchaseFrequency", "AvgSpend", "PreferredGenre", "TotalSpend")], centers = 3)
>
> # Add the cluster assignment to the original dataset
> customer_data$Cluster <- as.factor(kmeans_result$cluster)
>
> # View the first few rows of the data with clusters
> head(customer_data)
```

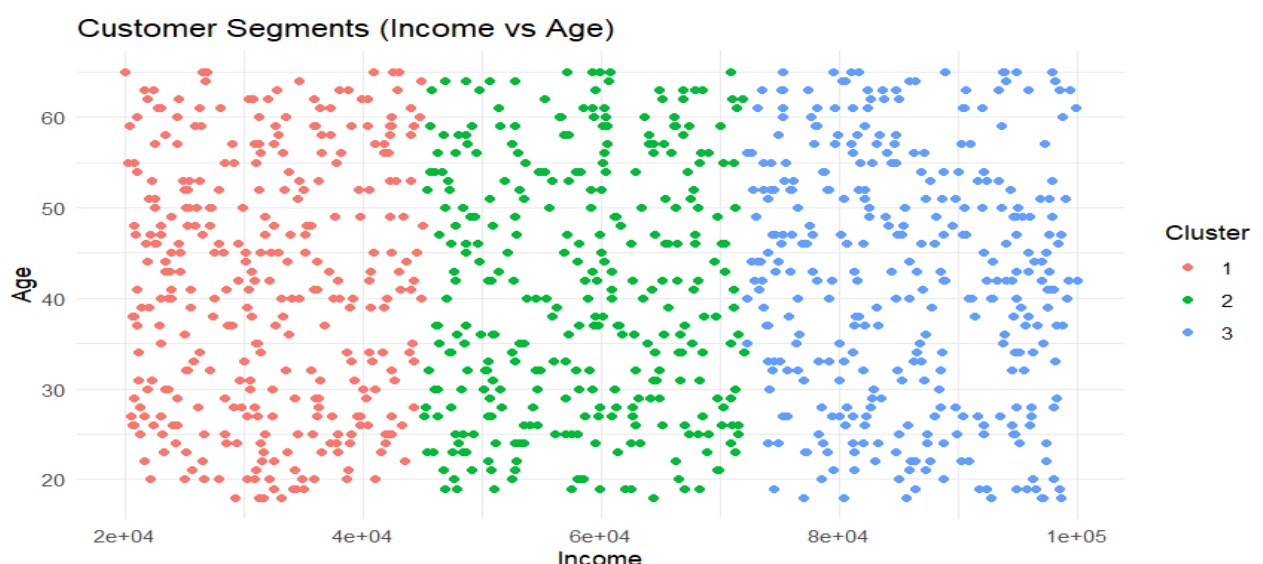
	CustomerID	Age	Income	PurchaseFrequency	AvgSpend	PreferredGenre	TotalSpend	Cluster
1	1	48	35523	8	29	Mystery	232	1
2	2	32	27052	17	10	Fiction	170	1
3	3	31	22221	5	13	Romance	65	1
4	4	20	40994	9	31	Sci-Fi	279	1
5	5	59	60458	10	13	Mystery	130	2
6	6	60	31165	13	31	Fiction	403	1

```
>
```

3. **Visualization & Interpretation:** Using the results of the clustering, we visualized the clusters and interpreted the characteristics of each segment, for this we used the libraries (ggplot2), (dplyr), (tidyr), library (GGally), library(ggcorrplot). The following sections summarize each cluster's key attributes and suggested marketing strategies.

Key Findings and Targeted Marketing Strategies

A. Customer Segments (Income vs. Age):



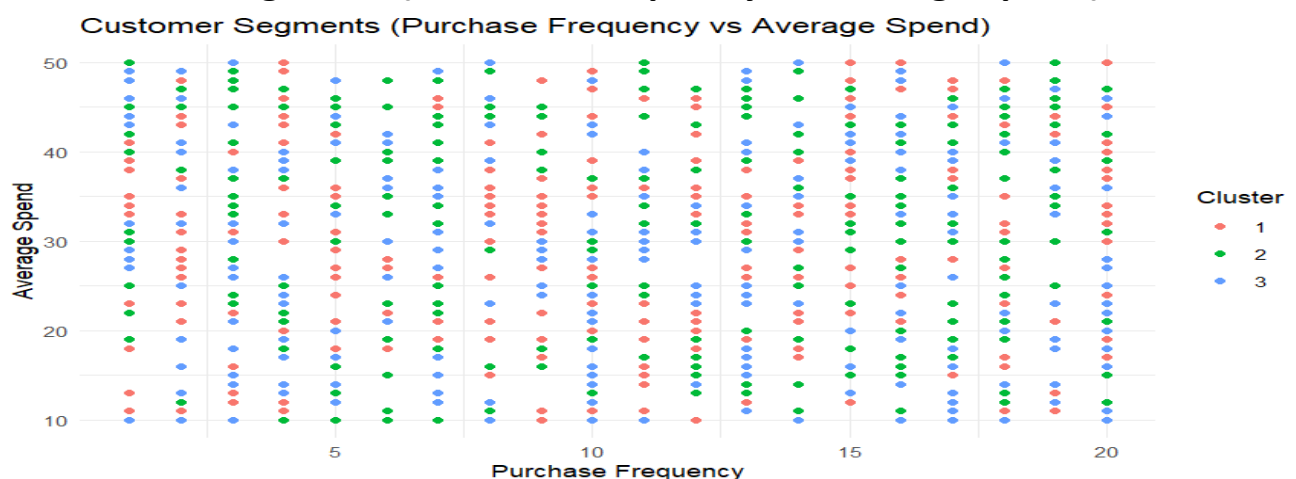
Interpretation: The scatter plot shows the distribution of customer segments based on **Income** and **Age**:

- **Cluster 1 (Red):** This segment is composed of younger customers with lower incomes, primarily positioned in the lower-left corner.
- **Cluster 2 (Green):** This cluster contains customers with a wider range of ages and income levels, located in the center.
- **Cluster 3 (Blue):** Older customers with higher incomes dominate this cluster, situated in the upper-right.

Targeted Marketing Strategies:

- **Cluster 1:** Focus on affordable product offerings, such as budget-friendly options or discounts for younger audiences. Use social media and influencer marketing to appeal to this demographic.
- **Cluster 2:** Develop a balance of affordable and mid-range products and target them through email campaigns or loyalty programs that emphasize value and practicality.
- **Cluster 3:** Market premium products, exclusive offers, and services that appeal to older, higher-income customers. Personalized communication through direct mail or premium newsletters could be effective.

B. Customer Segments (Purchase Frequency vs. Average Spend):



Interpretation: This scatter plot shows the relationship between Purchase Frequency and Average Spend:

- **Cluster 1 (Red):** Customers here make fewer purchases and spend less on average.
- **Cluster 2 (Green):** This group shows moderate purchase frequency and average spending.
- **Cluster 3 (Blue):** Customers in this segment are high spenders who purchase more frequently.

Targeted Marketing Strategies:

- **Cluster 1:** Focus on increasing customer engagement through targeted promotions or loyalty programs. Offer incentives such as discounts on the next purchase or bundling deals to encourage repeat purchases.
- **Cluster 2:** Engage this group with special offers to increase both purchase frequency and average spend. Highlight products that balance quality and cost.
- **Cluster 3:** Tailor marketing messages to this high-value segment, offering exclusive deals, VIP programs, or premium services. Emphasize loyalty rewards and personalized experiences to maintain their engagement.

C. Income Distribution by Customer Segment:



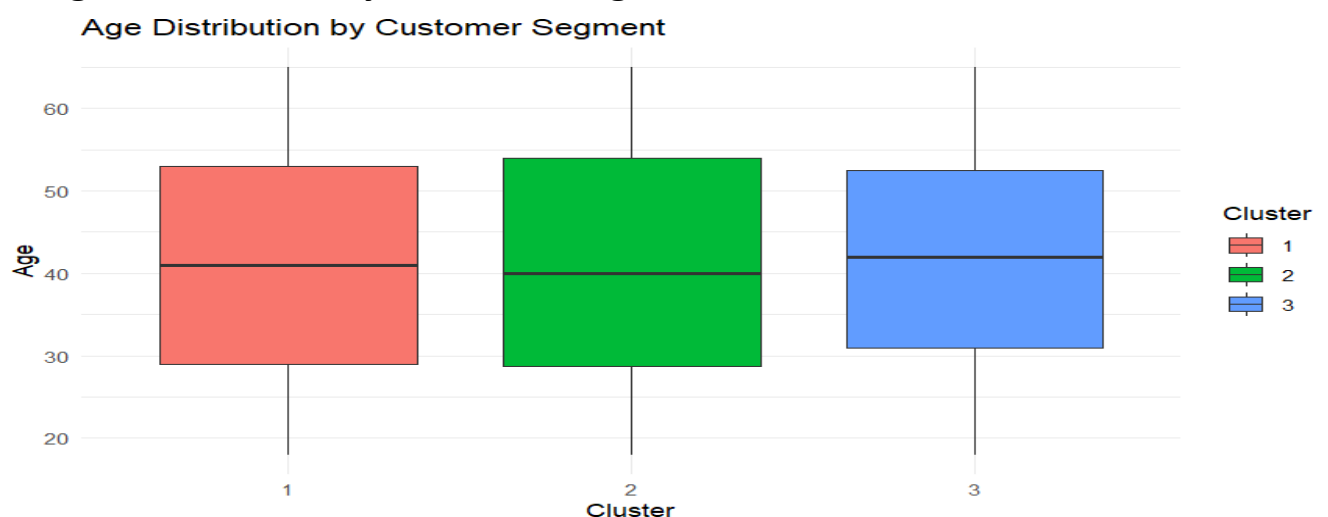
Interpretation: The box plot reveals **Income Distribution** across customer segments:

- **Cluster 1 (Red):** This segment has the lowest median income and a narrow range of values.
- **Cluster 2 (Green):** Represents a higher median income and a broader distribution of income.
- **Cluster 3 (Blue):** This group shows the highest median income, with a wider spread of income values.

Targeted Marketing Strategies:

- **Cluster 1:** Offer budget-conscious promotions or entry-level products that cater to younger, lower-income customers. Focus on price-sensitive messaging.
- **Cluster 2:** Develop mid-range product offerings that appeal to the varying income levels of this segment. Market them through discount events and product bundles.
- **Cluster 3:** Focus on high-end offerings and exclusive product launches. Highlight quality, luxury, and premium experiences that align with their higher income.

D. Age Distribution by Customer Segment:



Interpretation: The box plot shows **Age Distribution** for each cluster:

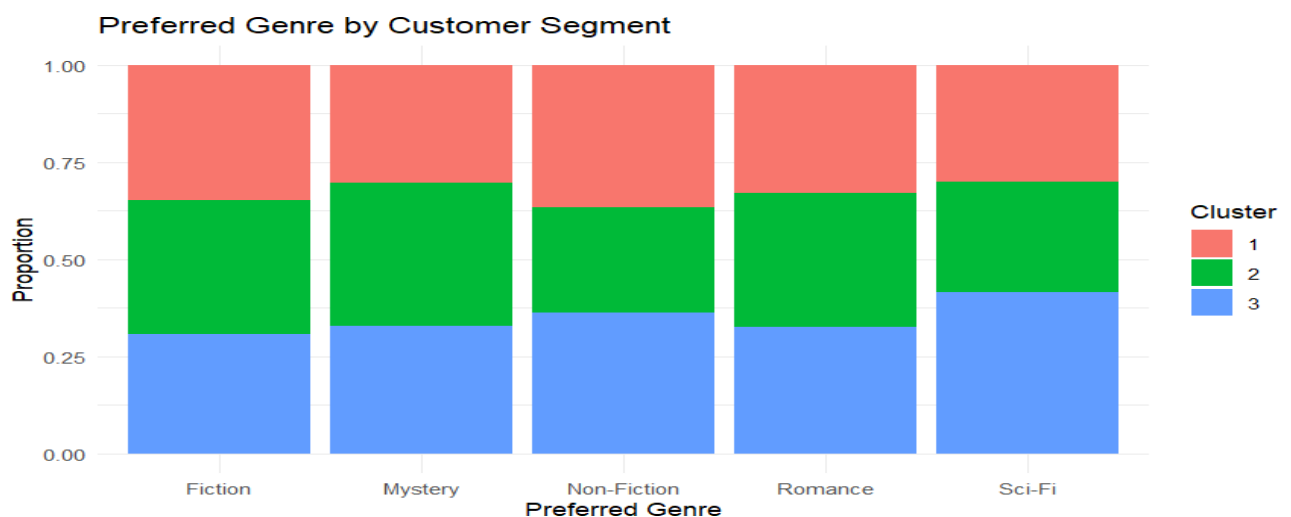
- **Cluster 1 (Red):** Primarily younger customers, with a narrow age range.

- **Cluster 2 (Green):** Slightly older customers with a broader age range.
- **Cluster 3 (Blue):** The age range is similar to Cluster 1 but slightly broader.

Targeted Marketing Strategies:

- **Cluster 1:** Engage with younger customers through digital channels like social media, highlighting trendy, affordable products.
- **Cluster 2:** Focus on products that cater to both practicality and quality. Utilize both traditional and digital marketing, balancing value and modernity in messaging.
- **Cluster 3:** Create campaigns emphasizing reliability, longevity, and experiences that appeal to older customers. Provide content that highlights the quality and sophistication of your products.

E. Preferred Genre by Customer Segment:



Interpretation: The bar chart shows customer preferences for book genres:

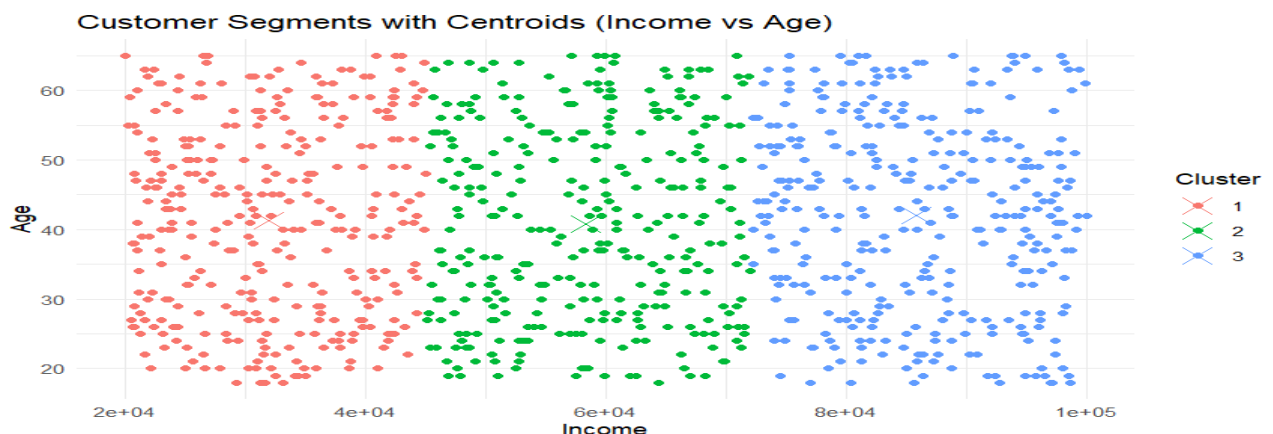
- **Cluster 1 (Red):** A higher proportion prefers Fiction, followed by Mystery and Non-Fiction.

- **Cluster 3 (Blue):** A notable preference for Sci-Fi among this cluster.

Targeted Marketing Strategies:

- **Cluster 1:** Focus on offering Fiction books with promotions that target this genre, such as bundle deals or discounts on the most popular titles.
- **Cluster 2:** Provide a balanced selection of genres, with special offers across a range of preferences, including Fiction, Non-Fiction, and Romance.
- **Cluster 3:** Market Sci-Fi books and exclusive content, such as limited editions or pre-release offers, to appeal to their specific interests.

F. Centroids of Customer Segments (Income vs. Age):



Interpretation: The scatter plot with centroids shows the average data points of each segment:

- **Cluster 1 (Red):** Younger customers with lower incomes.
- **Cluster 2 (Green):** A middle ground of both age and income.
- **Cluster 3 (Blue):** Older customers with higher incomes.

Targeted Marketing Strategies:

- **Cluster 1:** Position products as affordable and trendy, with a focus on younger customers' lifestyle needs.

- **Cluster 2:** Highlight products that provide a balance of quality and price, leveraging both online and offline marketing strategies to reach this varied demographic.
- **Cluster 3:** Target this cluster with exclusive, high-quality products. Focus on creating a premium experience through personalized customer service and tailored offers.

This structure focuses on **interpreting** each segment in terms of their demographic and behavioural characteristics and **suggesting targeted marketing strategies** accordingly. By following this format, your report can guide marketing teams in focusing their efforts on the most effective strategies for each cluster.

4. Final Thoughts

The segmentation analysis offers valuable insights into the diversity of our customer base, highlighting distinct groups that exhibit unique behaviours, preferences, and spending patterns. By understanding these segments, we can develop more effective, targeted marketing strategies that resonate with each group's specific needs and desires.

Key takeaways from the analysis include:

- **Younger, lower-income customers** prefer affordable and trendy products, making them ideal targets for budget promotions and influencer-driven campaigns.
- **Middle-aged customers** present opportunities for mid-range offerings, requiring a balance between quality and price in our messaging.
- **Older, higher-income customers** seek premium products and personalized experiences, offering a chance to engage them through exclusivity and loyalty programs.

These segments provide clear direction for allocating marketing resources, refining product offerings, and tailoring promotional

efforts. By prioritizing customer-centric strategies and focusing on each segment's preferences, we can maximize customer satisfaction and drive long-term business growth.

In conclusion, this analysis not only empowers us to connect with our customers more meaningfully but also positions the company for greater success through personalized and impactful marketing strategies.

2. Conjoint Analysis

1. Objective:

The purpose of this analysis is to perform a **Conjoint Analysis** to assess how different attributes, such as **Preferred Genre** and **Average Spend**, influence customer preferences. The insights gained from this analysis will help in better understanding customer behavior and designing products or services that align with their preferences.

2. Data Overview:

For this analysis we used the previously generated dataset **Book Store** Datast in the Customer Segmentation. The dataset used in this analysis is based on customer information related to their **Preferred Genre, Average Spend, Age, Income**, and other characteristics. To the Conjoint Analysis, we focused on **Preferred Genre** and **Average Spend** as the key attributes. These attributes were considered because they have significant implications in designing targeted marketing strategies.

The dataset contains customer preferences for various combinations of attribute levels (i.e., different **Genres** and **Spending Levels**). A random preference rating was used for each profile to simulate the choices customers might make based on these attributes.

3. Methodology:

To perform the conjoint analysis, the following steps were taken:

1. Data Preparation:

- **Attributes and Levels:** The two attributes considered were:
 - **Preferred Genre:** Mystery, Fiction, Romance, Sci-Fi, Non-Fiction.
 - **Average Spend:** Low, Medium, High.

Attributes Output:

```
> # Print the attributes to verify
> print(attributes)
$PreferredGenre
[1] "Mystery"      "Fiction"      "Romance"      "Sci-Fi"      "Non-Fiction"

$AvgSpend
 [1] Medium Low      Low      Medium Low      Medium Medium Medium High      Low      Low      Medium
[13] Medium Medium High      High      Medium Medium Low      High      High      Medium Low      High
[25] Low      High      High      Low      High      Medium High      Medium High      Low      Medium High
[37] Medium Medium Medium Medium Medium Medium High      High      Medium High      Medium High
[49] Low      Medium Low      High      Low      High      High      Low      High      Low      High      High
[61] Low      High      Medium Low      Medium High      Low      High      Medium Medium Low      Low
[73] High      High      Medium High      Low      Low      Medium Low      High      High      Low      High
[85] Low      Medium Low      High      Medium Medium Medium Low      Medium High      Low      Medium
```

2. Profile Matrix:

- A profile matrix was created by combining all possible combinations of the attribute levels (genres and spend levels).

3. Preference Simulation:

- Random preference ratings were assigned to each profile to simulate how customers might rank their preferences for each combination of attributes.

4. Conjoint Model:

- The **caModel()** function from the conjoint package was used to perform the Conjoint Analysis, generating the **part-worth utilities** (coefficients) for each attribute and its corresponding levels.

Conjoint Analysis Output:

```
> # Print the results
> summary(ca_result)
      Length Class  Mode
call          2 -none- call
terms         3  terms  call
residuals    5000 -none- numeric
coefficients  28 -none- numeric
aliased       7 -none- logical
sigma         1 -none- numeric
df            3 -none- numeric
r.squared     1 -none- numeric
adj.r.squared 1 -none- numeric
fstatistic    3 -none- numeric
cov.unscaled  49 -none- numeric
>
```

Conjoint Analysis Coefficients Results:

```
> # Extract and print the coefficients (part-worth utilities)
> coefficients <- ca_result$coefficients
> print(coefficients)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.990072480	0.01631313	183.29239799	0.0000000
factor(x\$PreferredGenre)1	0.050566027	0.03261517	1.55038354	0.1211129
factor(x\$PreferredGenre)2	-0.029986507	0.03261517	-0.91940359	0.3579289
factor(x\$PreferredGenre)3	-0.045341659	0.03261517	-1.39020141	0.1645297
factor(x\$PreferredGenre)4	-0.009614725	0.03261517	-0.29479301	0.7681643
factor(x\$AvgSpend)1	-0.001144789	0.02310693	-0.04954309	0.9604885
factor(x\$AvgSpend)2	-0.012497779	0.02287056	-0.54645709	0.5847762

5. Interpretation of Results:

- The part-worth utilities (coefficients) were examined to determine the relative importance of each attribute in shaping customer preferences.

4. Results and Interpretation:

a. Part-Worth Utilities:

The part-worth utilities were extracted from the model to understand the impact of each attribute level on customer preferences. The results are shown in the following table:

Output:

```
> # Combine the intercept with the rest
> df <- rbind(intercept, df)
>
> # Print the data frame to verify
> print(df)
```

	Attribute	Level	Utility
(Intercept)	Intercept	Intercept	2.990072480
factor(x\$PreferredGenre)1	PreferredGenre	Medium	0.050566027
factor(x\$PreferredGenre)2	PreferredGenre	High	-0.029986507
factor(x\$PreferredGenre)3	PreferredGenre	Medium	-0.045341659
factor(x\$PreferredGenre)4	PreferredGenre	High	-0.009614725
factor(x\$AvgSpend)1	AvgSpend	Medium	-0.001144789
factor(x\$AvgSpend)2	AvgSpend	High	-0.012497779

Attribute	Level	Utility
Intercept	Intercept	2.99
Preferred Genre	Medium	0.05
Preferred Genre	High	-0.03
AvgSpend	Medium	-0.001
AvgSpend	High	-0.01

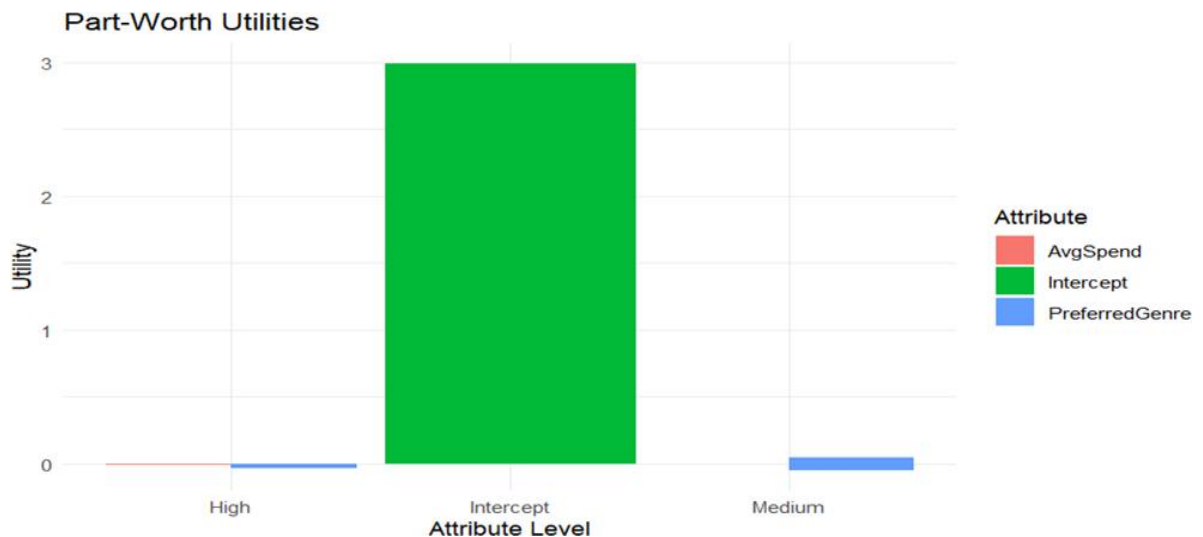
- **Intercept (2.99):** The highest utility value, representing the baseline or overall preference level.
- **Preferred Genre (Medium):** Positive utility of 0.05 suggests a slight preference for medium genres.
- **Preferred Genre (High):** A negative utility of -0.03 indicates that "High" genres have a lower utility compared to the baseline genre.
- **AvgSpend (Medium and High):** Both "Medium" and "High" spending levels show minimal influence on preference, with values close to zero. This suggests that customers' average

spending habits have a relatively low impact on their genre preferences.

b. Visualization:

Several visualizations were created to better understand the results.

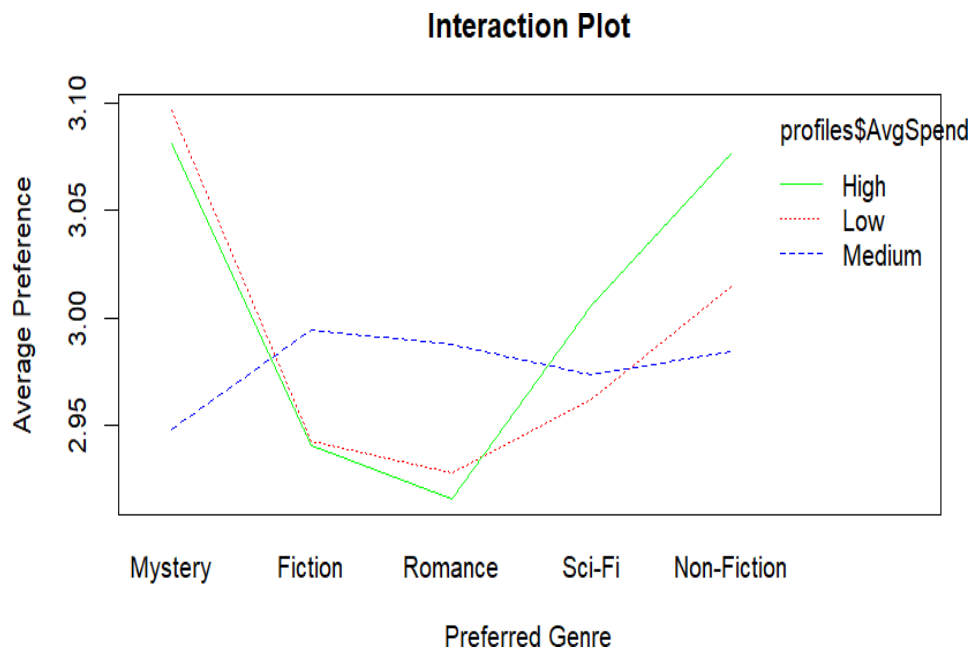
1. Bar Plot of Part-Worth Utilities:



- This plot illustrates the utility of different attribute levels. The chart shows that the **Intercept** has the highest utility, followed by **Preferred Genre (Medium)**. The **AvgSpend** levels have the least influence on preferences.

Interpretation: The **Intercept** is the most critical factor affecting customer preferences, which could represent a baseline preference. The **Preferred Genre** attributes play a more prominent role than **AvgSpend** in shaping customer behaviour.

2. Interaction Plot for Preferred Genre and AvgSpend:



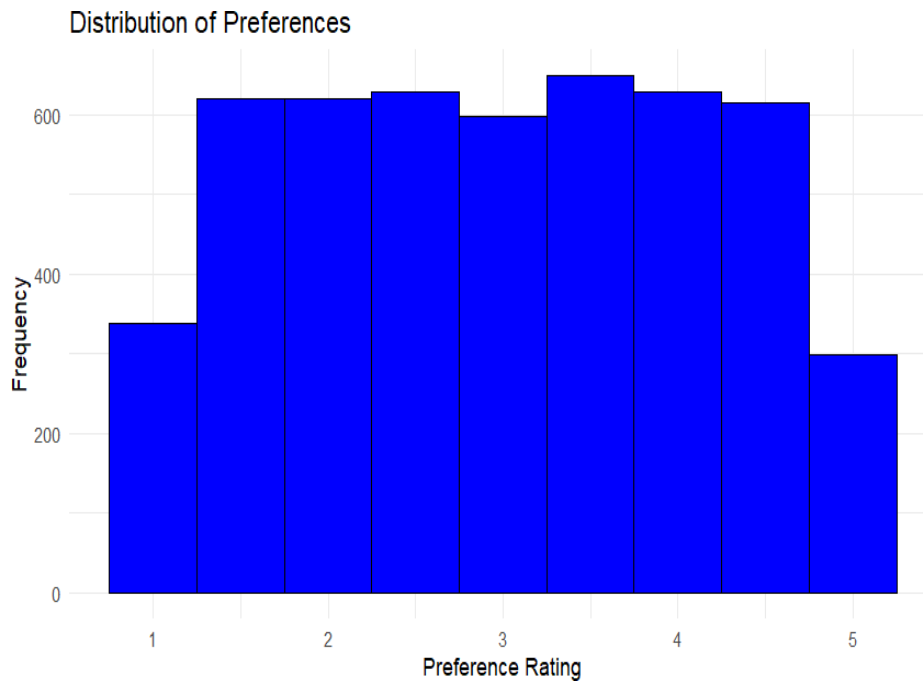
- This plot visualizes how **Preferred Genre** and **AvgSpend** interact with each other to influence customer preferences. It shows how preference ratings for different genres change across various spend levels.

Interpretation:

- For **High Spenders**, preferences steadily increase from **Mystery** to **Non-Fiction**.
- For **Low Spenders**, preferences shift differently, with a higher preference for **Fiction** than **High Spenders**.
- **Medium Spenders** show a varied pattern, with a peak preference for **Romance**.

Interaction Effect: The non-parallel lines indicate an interaction effect between **Preferred Genre** and **AvgSpend**, suggesting that the relationship between genre preferences and spending is not uniform across all spending levels.

3. Histogram of Preferences:



- The histogram displays the distribution of **preference ratings**, which range from 1 to 5.

Interpretation:

- The distribution is approximately **normal**, with most ratings concentrated around the middle values (3).
- This suggests that the majority of customers have **moderate preferences**, with fewer customers expressing extreme preferences (either low or high).

5. Recommendations:

Based on the conjoint analysis and interpretation of the results, the following recommendations can be made:

1. Product Design:

- Since **Preferred Genre** has a higher utility than **AvgSpend**, product offerings should focus more on catering to different genre preferences rather than pricing alone.
- Genres such as **Non-Fiction** seem to be favoured, and future product designs could align more with this genre.

2. Targeted Marketing:

- **Medium and High Spenders** show less sensitivity to spending levels, but they do exhibit distinct preferences for genres. Therefore, marketing efforts should emphasize genre-related content.
- For **Low Spenders**, emphasize **Fiction** genres in promotions and tailor offers that might appeal to this group.

3. Customer Segmentation:

- Use **genre preferences** as the primary segmentation criterion in marketing campaigns. Focus on high-preference genres for each segment to improve engagement.

6. Conclusion:

The conjoint analysis reveals that **Preferred Genre** plays a more prominent role in shaping customer preferences than **Average Spend**. The most significant takeaway from the analysis is that **product offerings should prioritize customer preferences in terms of genre**, with particular attention to **Non-Fiction** as a popular genre. The insights also emphasize the need for **targeted marketing** and **personalized product recommendations** based on customer behaviour.

By leveraging these insights, the business can create more personalized offerings that resonate with different customer segments, leading to increased customer satisfaction and business growth.

3. Google Analytics

1. Audience Overview:

A. User Attributes Overview Dashboard:



Key Insights:

- **Active Users by Country:** A world map visualizes the distribution of active users across various countries. The United States has the highest number of active users, followed by Canada and India.
- **Active Users in Last 30 Minutes:** A bar chart shows the number of active users in the last 30 minutes. The data fluctuates, indicating user activity patterns.
- **Active Users per Minute:** A line graph illustrates the trend of active users per minute over time, identifying peak usage periods.
- **Top Countries:** A table lists the top countries with the highest number of active users.
- **Active Users by City:** A table ranks cities based on their number of active users. Aspen has the highest number, followed by New York and Brook Park.

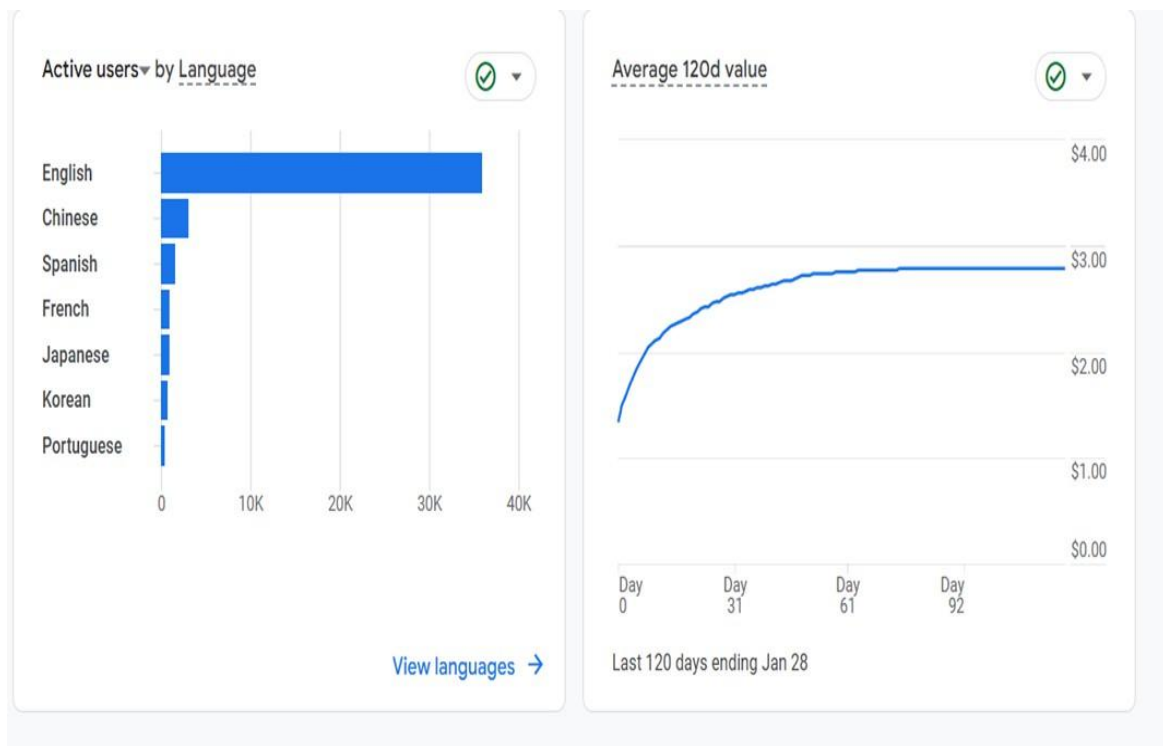
B. User Attributes Overview Dashboard:



Key Insights:

- **Active Users by Gender:** A pie chart visualizes the distribution of active users between male and female. 61.1% are male, 38.9% are female.
- **Active Users by Interests:** A bar chart shows the number of active users based on their interests, with top interests being Technology/Technophiles, News & Politics/Avid News, and Banking & Finance/Avid Investors.
- **Active Users by Age:** A bar chart shows the number of active users based on their age, with the majority between 18-24, followed by 25-34 and 35-44.

C. User Attributes Overview Dashboard



Key Insights:

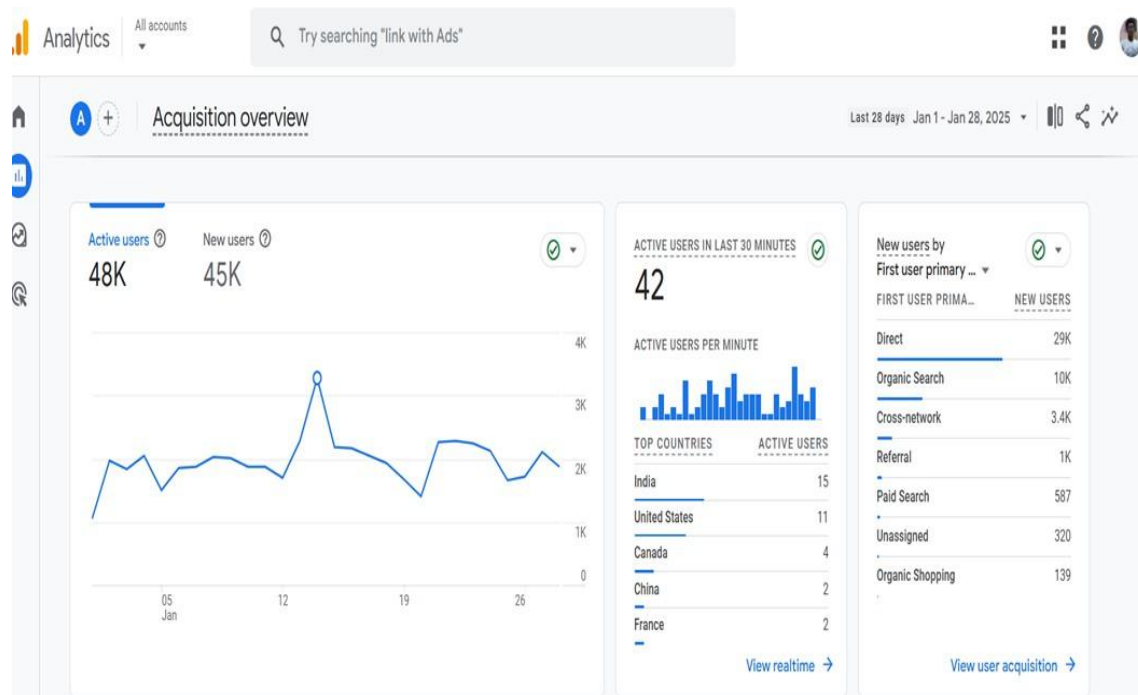
- **Active Users by Language:** A bar chart visualizes the distribution of active users across different languages. English has the highest number, followed by Chinese and Spanish.
- **Average 120d Value:** A line graph shows the average value over the last 120 days, indicating a steady increase.

Why and How These Analytics are Used:

- These analytics provide a snapshot of user activity and distribution, helping in understanding user behaviour and identifying key markets or regions. Marketers and business strategists can use this data to target campaigns more effectively and allocate resources to high-performing areas.
- Understanding user demographics, interests, and languages allows businesses to tailor their content and marketing strategies to better suit their audience, improving engagement and conversion rates.

2. Acquisition Overview:

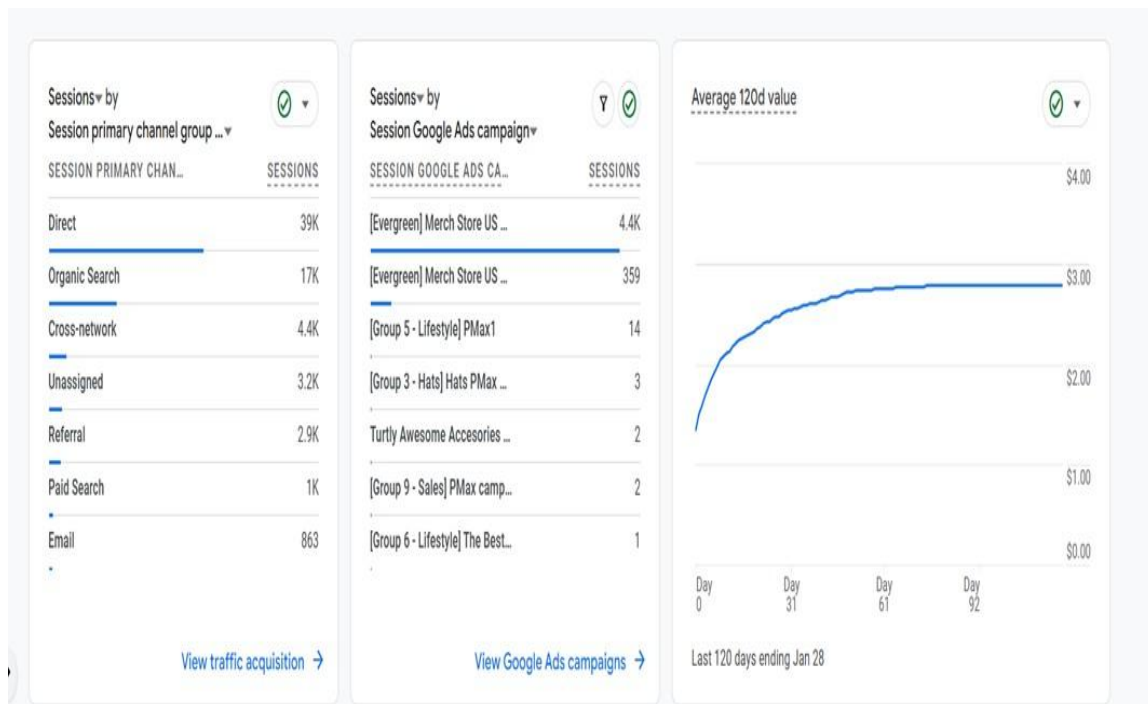
A. Acquisition Data Overview Dashboard:



Key Insights:

- **Active Users:** 48K active users within the specified timeframe, with a line graph showing trends over the last 28 days.
- **New Users:** 45K new users acquired during the same timeframe.
- **Active Users in Last 30 Minutes:** 42 active users in the last 30 minutes, providing a real-time snapshot.
- **Top Countries:** India leads with 15 active users, followed by the United States and Canada.
- **New Users by First User Primary Source:** Direct traffic accounts for the highest percentage of new users, followed by Organic Search and Cross-network.

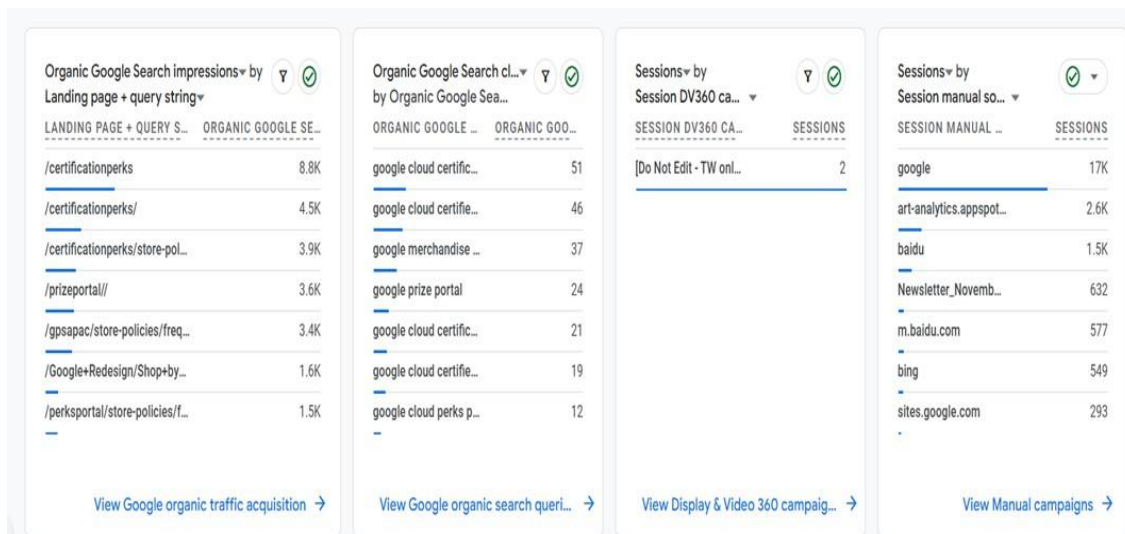
B. Acquisition Data Overview Dashboard:



Key Insights:

- **Sessions by Session Primary Channel Group:** Direct traffic accounts for the highest number of sessions, followed by Organic Search and Cross-network.
- **Sessions by Session Google Ads Campaign:** The top-performing campaign is "[Evergreen] Merch Store US."
- **Average 120d Value:** The average value has shown a steady increase over the last 120 days.

C. Acquisition Data Overview Dashboard:



Key Insights:

- **Organic Google Search Impressions by Landing Page + Query String:** The top landing page + query string combination is "/certification perks" and "google cloud certification."
- **Organic Google Search Clicks by Organic Google Search Query:** The top query string is "google cloud certification."
- **Sessions by Session DV360 Campaign:** The top-performing campaign is "[Do Not Edit-TW only]."
- **Sessions by Session Manual Source:** The top manual source is "google."

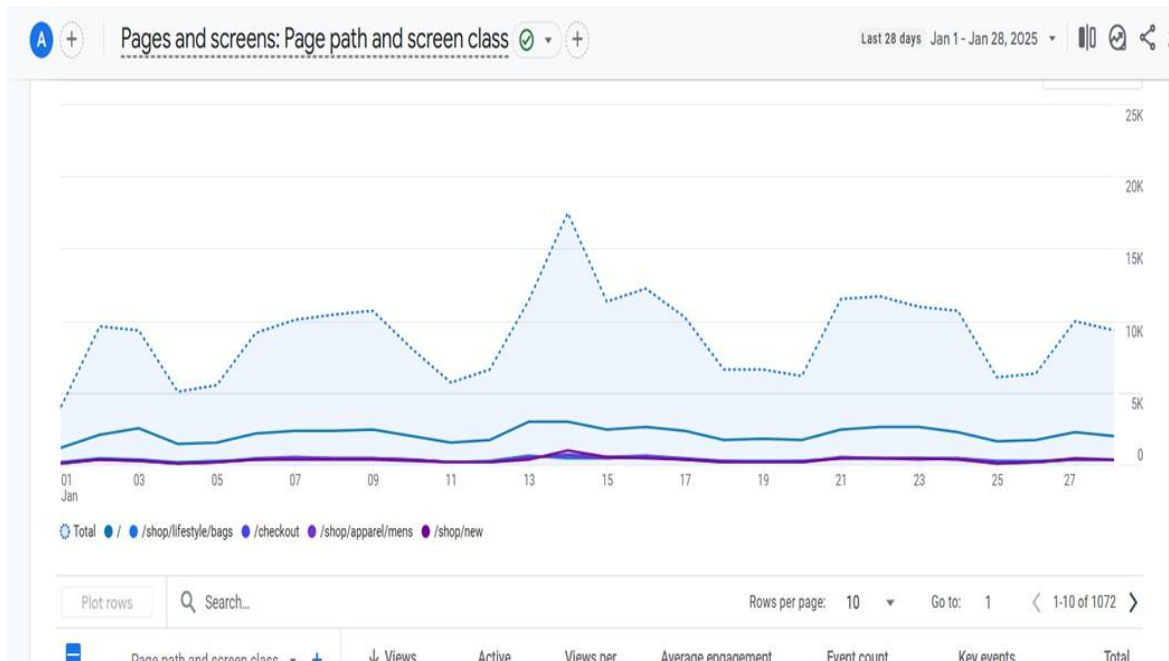
Why and How These Analytics are Used:

- Acquisition data helps in understanding how users find your site, allowing marketers to optimize channels that drive the most traffic and adjust strategies for those that are underperforming.
- Identifying the most effective channels and campaigns helps allocate marketing budgets more efficiently, ensuring maximum return on investment.

- Analysing search impressions and clicks helps in optimizing SEO efforts, while understanding session sources from campaigns aids in refining Ad strategies.

3. Behaviour Flow:

A. Page Path and Screen Class Data Overview:



Key Insights:

- Page Path and Screen Class:** A line graph visualizes the number of views for different page paths and screen classes over the last 28 days.
- Total Views:** This represents the total number of views across all page paths and screen classes.

B. Page Paths and Screen Classes Table:

Plot rows		Search...						
Page path and screen class		Views	Active users	Views per active user	Average engagement time per active user	Event count All events	Key events All events	Total revenue
☒	Total	252,734 100% of total	48,067 100% of total	5.26 Avg 0%	1m 18s Avg 0%	964,902 100% of total	61,022.00 100% of total	\$113,736.39 100% of total
☒	1 /	59,592	33,923	1.76	19s	227,488	9.00	\$0.00
☒	2 /shop/lifestyle/bags	10,269	5,903	1.74	28s	41,223	1.00	\$0.00
☒	3 /checkout	10,023	2,264	4.43	1m 58s	27,525	786.00	\$110,947.84
☒	4 /shop/apparel/mens	9,226	4,076	2.26	1m 06s	46,459	4.00	\$0.00
☒	5 /shop/new	9,043	4,689	1.93	54s	41,216	2.00	\$0.00
☐	6 /shop/apparel	8,187	4,711	1.74	51s	42,243	3.00	\$0.00
☐	7 /shop/clearance	7,583	4,147	1.83	34s	27,219	2.00	\$0.00
☐	8 /search	7,357	2,832	2.60	45s	16,815	2.00	\$0.00
☐	9 /shop/lifestyle/drinkware	5,862	3,324	1.76	38s	25,697	1.00	\$0.00
☐	10 /canada	4,864	2,915	1.67	10s	11,746	0.00	\$0.00

Key Insights:

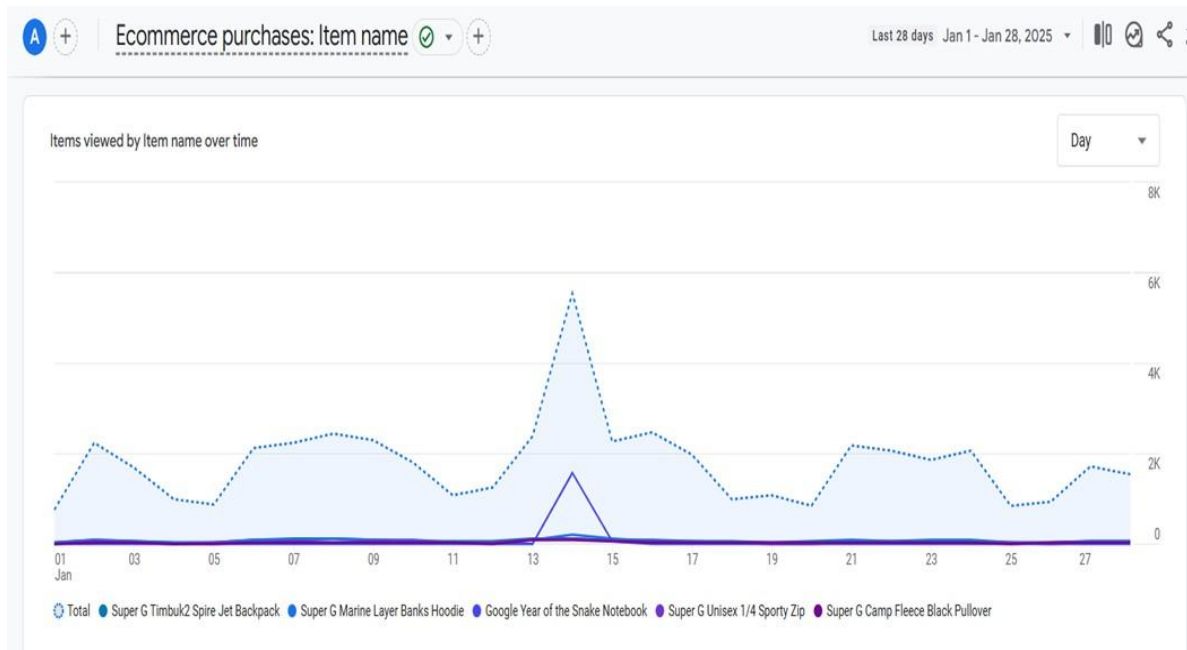
- **Views per Active User:** The /checkout page path has the highest views per active user (4.43) and the highest average engagement time (1m 58s).
- **Total Revenue:** The /checkout page path also generated the most revenue (\$110,947.84).
- **Active Users:** The /shop/new page path has the highest number of active users (4,689).

Why and How These Analytics are Used:

- Behaviour flow analysis helps in understanding how users navigate through the site, identifying any drop-off points or popular paths. This can inform website design improvements and content optimization.
- Understanding user engagement and revenue generation on specific pages helps prioritize content and feature development, ensuring a focus on high-impact areas.

4. Conversion Goals:

A. Ecommerce Purchases by Item Name Overview:



Key Insights:

- **Items Viewed by Item Name Over Time:** The Super G Timbuk2 Spire Jet Backpack has the highest number of views.

B. Ecommerce Purchases Table:

Ecommerce purchases: Item name		Last 28 days Jan 1 - Jan 28, 2025			
Plot rows		Rows per page: 10 Go to: 1 1-10 of 406			
Item name		Items viewed	Items added to cart	Items purchased	Item revenue
Total		50,335 100% of total	26,410 100% of total	9,934 100% of total	\$153,301.90 100% of total
1	Super G Timbuk2 Spire Jet Backpack	1,898	435	37	\$3,335.40
2	Super G Marine Layer Banks Hoodie	1,748	190	32	\$3,225.00
3	Google Year of the Snake Notebook	1,603	103	50	\$888.80
4	Super G Unisex 1/4 Sporty Zip	956	41	8	\$469.20
5	Super G Camp Fleece Black Pullover	859	85	68	\$5,244.00
6	Android Classic Plushie	730	875	240	\$6,940.80
7	Google Timbuk2 Eco Nautical Belt Bag	661	110	39	\$1,548.40
8	Google Black Eco Zip Hoodie	625	45	6	\$331.20
9	Google Harbor Backpack	615	113	22	\$1,116.00
10	Google Portsmouth 1/4 Zip Sweater	553	82	27	\$1,738.00

Key Insights:

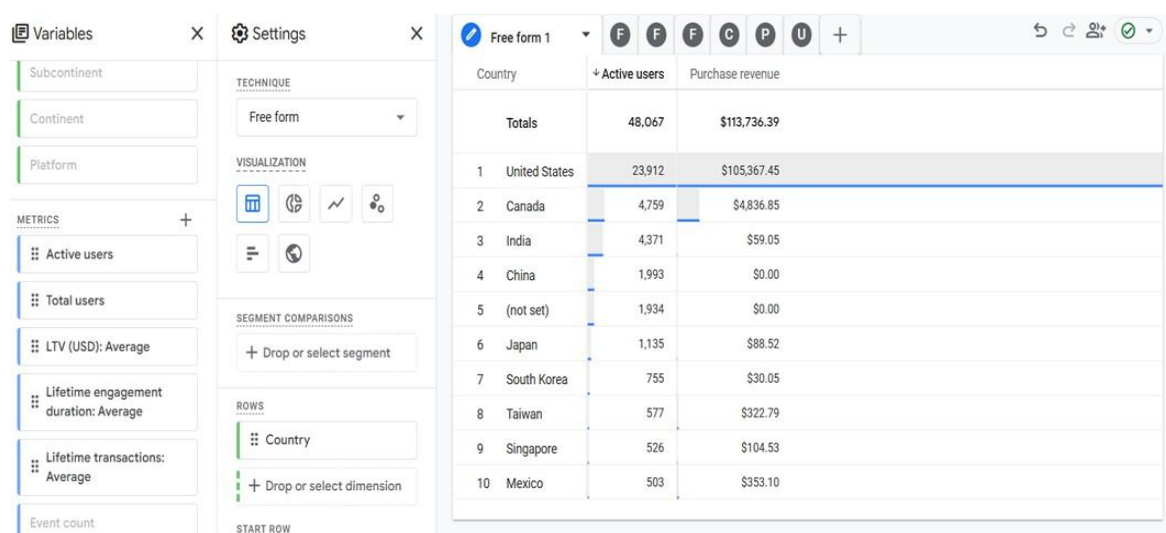
- **Most Popular Item:** The Super G Timbuk2 Spire Jet Backpack was the most popular item with 1,898 views and 37 purchases.
- **Most Added to Cart:** The Android Classic Plushie was the most added to cart, with 875 additions.
- **Highest Revenue:** The Android Classic Plushie generated the most revenue, with \$6,940.80.

Why and How These Analytics are Used:

- Conversion goals analytics track the performance of individual products, enabling businesses to identify best-sellers and optimize inventory and marketing strategies accordingly.
- Analysing ecommerce performance by item helps businesses understand customer preferences and optimize product offerings and pricing strategies.

5. Custom Report:

A. Custom Analysis Dashboard:

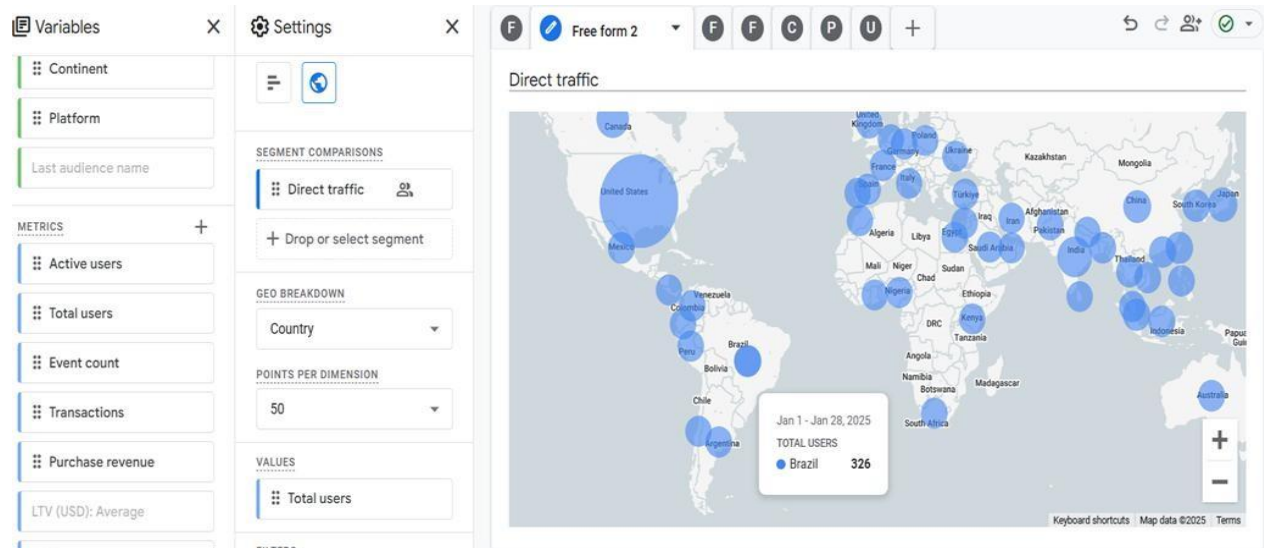


Key Insights:

- **Totals:** The total number of active users (48,067) and total purchase revenue (\$113,736.39).

- **United States:** Has the highest number of active users (23,912) and generates the most revenue (\$105,367.45).

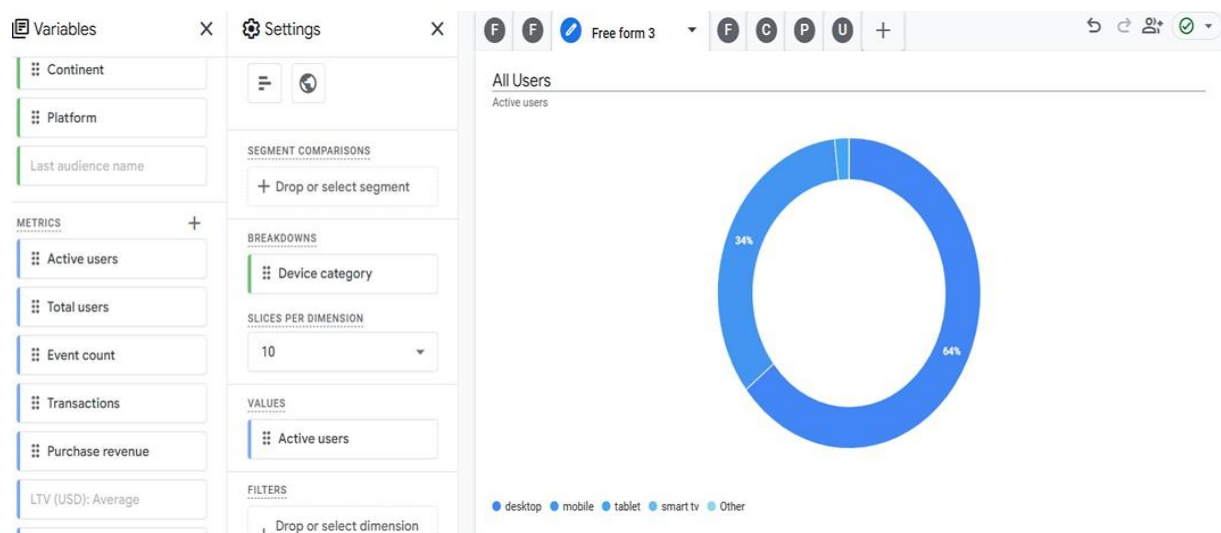
B. Map Visualization:



Key Insights:

- **Distribution of Total Users:** The United States has the highest number of total users, followed by Brazil, India, and Japan.

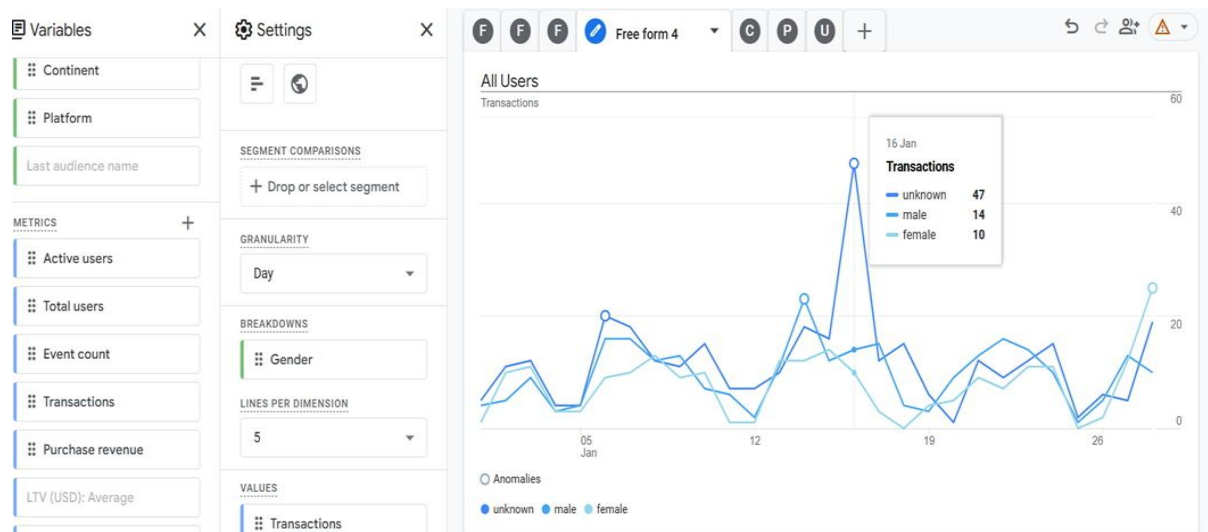
C. Device Category Distribution:



Key Insights:

- **Active Users by Device:** 64% of active users access the platform via desktop devices.

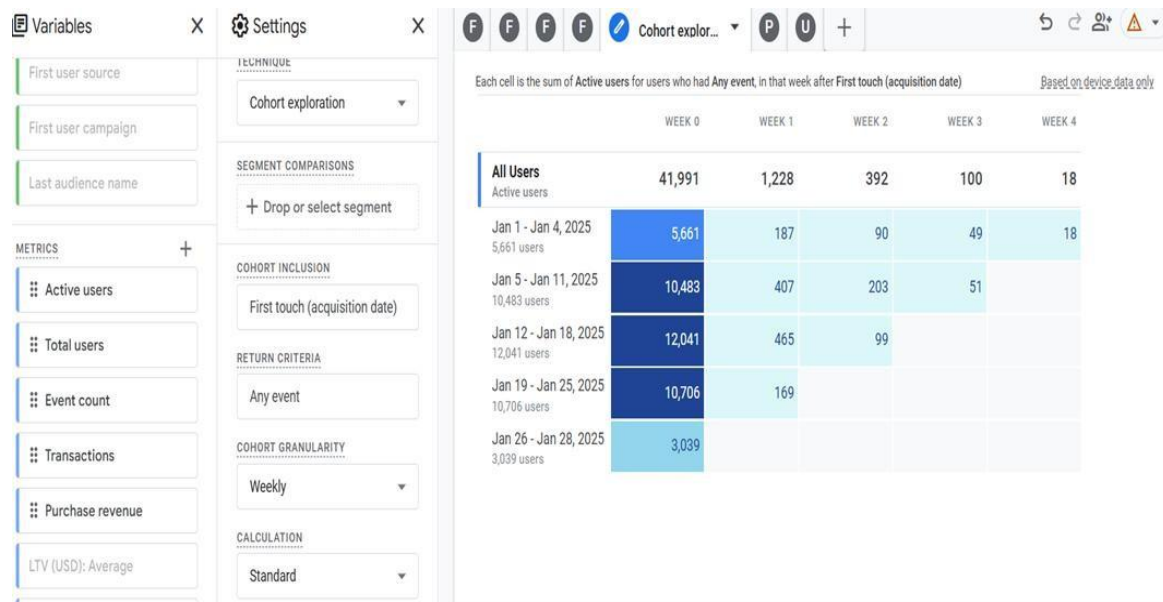
D. Transactions Trend by Gender:



Key Insights:

- **Trend of Transactions:** The total number of transactions fluctuates over time, with the number of transactions for "unknown" users consistently higher.

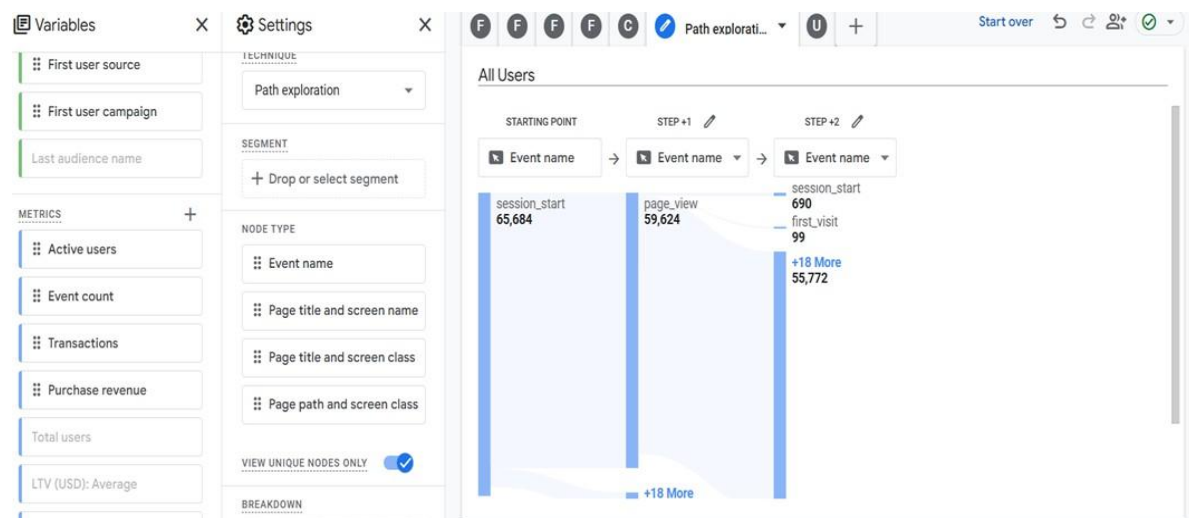
E. Cohort Analysis:



Key Insights:

- **Cohort Exploration:** The first cohort (Jan 1-4) has the highest number of initial users (5,661).

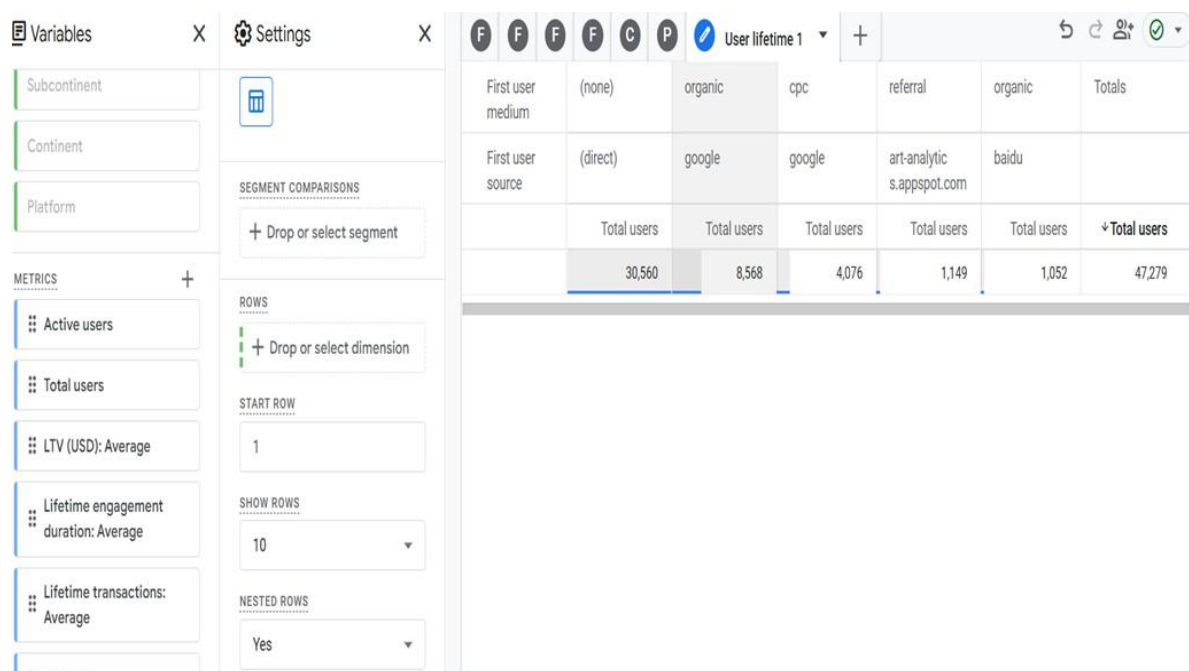
F. Path Exploration Visualization:



Key Insights:

- **User Journey:** The starting point is the "session_start" event, followed by "page_view" and "first_visit."

G. User Lifetime Analysis:



Key Insights:

- **First User Medium:** "Organic" traffic has the highest number of total users (30,560).

Why and How These Analytics are Used:

- Custom reports allow businesses to tailor analysis to specific needs, providing insights that are directly applicable to strategic decision-making.
- Geographic analysis helps businesses identify key markets and tailor marketing efforts to specific regions.
- Understanding device preferences allows businesses to optimize their websites and apps for the most popular devices, improving user experience.
- Analysing transaction trends by gender helps businesses understand purchasing behaviour, enabling targeted marketing and product development.
- Cohort analysis allows businesses to track user engagement and retention over time, providing insights into user lifecycle and the effectiveness of acquisition strategies.
- Path exploration helps businesses understand the typical user journey, identifying key steps and potential bottlenecks in the conversion process.

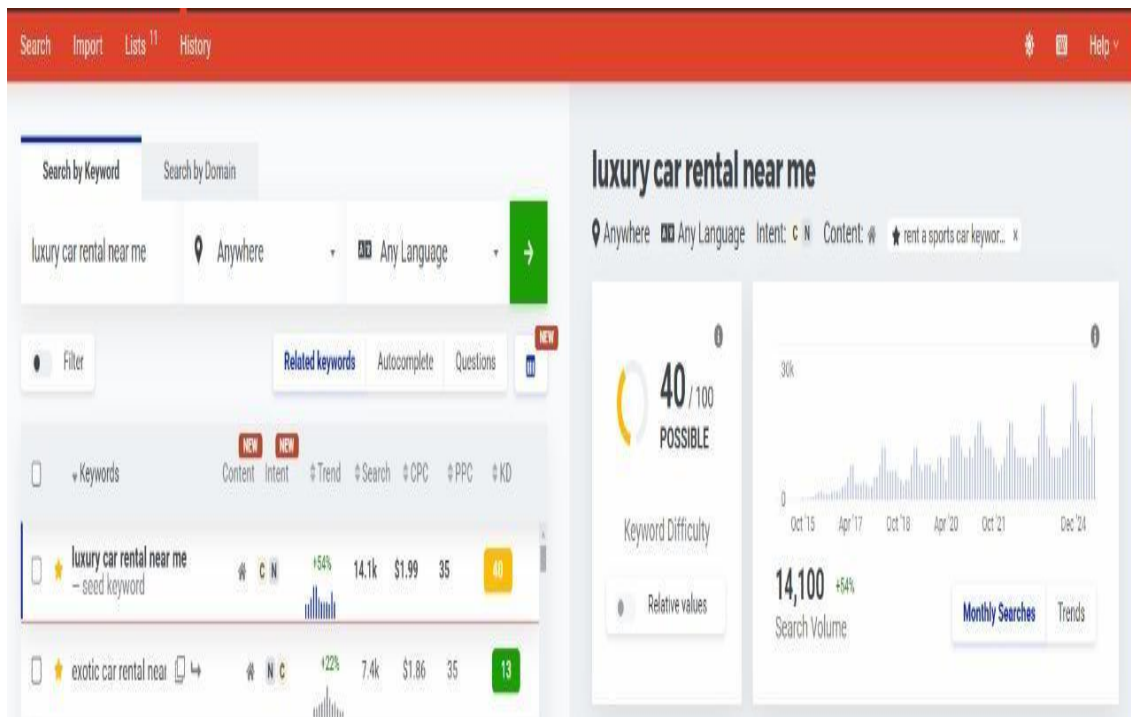
4.Keywords & Social Media Marketing

Objective:

To identify and justify two long-tail keywords for a luxury car rental business and create a corresponding social media advertisement, targeting potential customers on platforms like Facebook or Instagram.

1. Long-Tail Keyword Selection

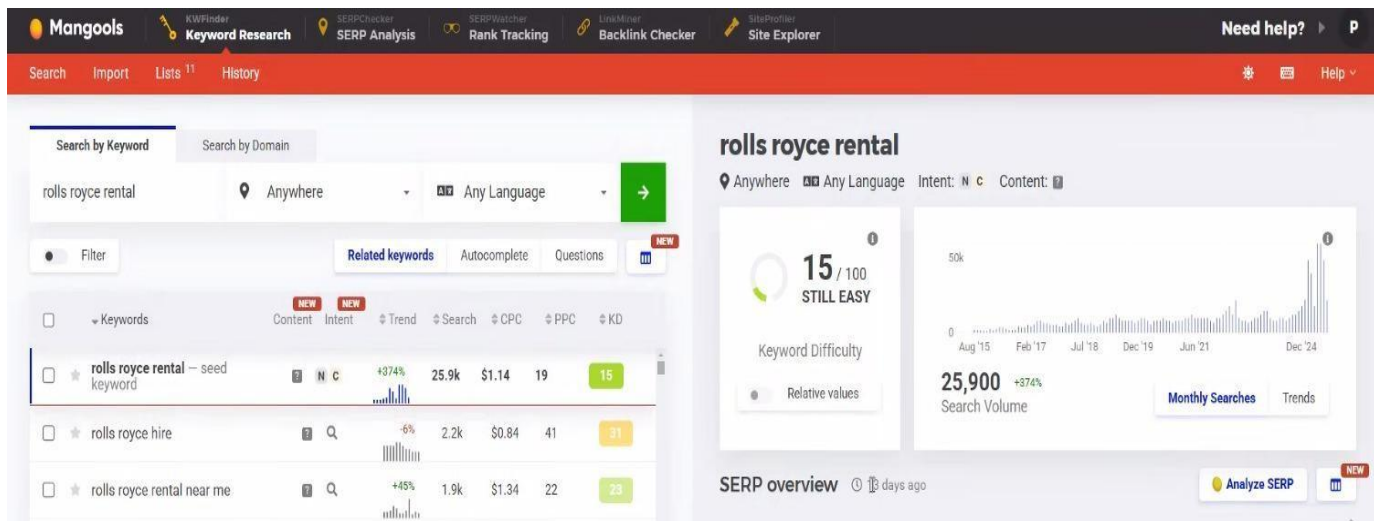
Keyword 1: "Luxury car rental near me"



- **Trend:** +54%
- **Search Volume:** 14,100
- **CPC:** \$1.99
- **PPC:** 35
- **Keyword Difficulty:** 40/100

Justification for Selection: This keyword was selected because it directly targets users who are actively looking for luxury car rentals nearby. With a search volume of 14,100, it demonstrates a solid interest in the topic. The trend of +54% suggests growing demand for this type of service. The CPC of \$1.99 is reasonable, and the keyword difficulty of 40/100 indicates moderate competition, which makes it a viable option for a targeted campaign. The relevance to the luxury car rental business is clear, as it focuses on geographical location and customer intent, increasing the likelihood of conversions.

Keyword 2: "Rolls Royce rental"



- **Trend:** +374%
- **Search Volume:** 25,900
- **CPC:** \$1.14
- **PPC:** 19
- **Keyword Difficulty:** 15/100

Justification for Selection: The keyword "Rolls Royce rental" was chosen due to its extremely high search volume of 25,900 and an impressive growth trend of +374%. Despite being a luxury brand, it still maintains a relatively low keyword difficulty of 15/100, indicating that it may be an easier target compared to other high-end vehicle rental keywords. The CPC of \$1.14 is quite affordable, especially for such a high-demand keyword. It is highly relevant to the business, targeting users specifically interested in renting one of the most prestigious car brands globally. This keyword is ideal for drawing in customers who are looking for premium, exclusive car rental services.

2. Social Media Ad Creation

Ad Platform: Instagram

Target Audience: High-income individuals, travel enthusiasts, and those seeking luxury experiences.

Ad Elements:

- **Headline:**

"BEST CAR RENT FOR YOUR TRIP"

- The headline emphasizes quality and convenience, positioning the service as the best choice for those looking for luxury car rentals.

- **Visual:**

The ad features a high-end luxury sports car (likely a McLaren GT), which captures the attention of affluent individuals looking for premium travel experiences. This visual appeals to customers who value luxury, performance, and exclusivity in their car rental choices.



- **Description:**

"Experience luxury like never before with our premium car rental services. Whether you're on a business trip, vacation, or simply looking for a luxury ride, we have the perfect car for your journey. Book now and enjoy a 60% discount!"

- This description highlights the main value proposition (luxury experience) and reinforces the time-sensitive nature of the offer with the 60% discount.
- **Call to Action (CTA):**
"BOOK NOW" & "MORE INFO"
 - The CTA buttons create a sense of urgency and encourage users to take immediate action or learn more about the service.
- **Discount Offer:**
"60% OFF"
 - The substantial discount makes the ad more attractive, enticing users to make a booking while the promotion lasts.
- **Contact Information:**
A phone number (123-456-7890) is provided for those who prefer to inquire or book by phone, offering an additional, personal channel for conversions.

3. Data Points to Support the Keyword Selection

- **Search Volume:**
 - "Luxury car rental near me" (14,100 monthly searches) shows consistent demand for rental services targeted to specific locations.
 - "Rolls Royce rental" (25,900 monthly searches) represents a niche but highly sought-after market in luxury car rentals.
- **CPC (Cost Per Click):**
 - Both keywords are relatively affordable considering their high search volume, making them effective for paid campaigns.
- **Keyword Difficulty:**

- "Luxury car rental near me" has a moderate difficulty level of 40/100, suggesting that competition is manageable for targeted efforts.
- "Rolls Royce rental" has a low difficulty of 15/100, indicating that it's an opportunity to target a high-volume, niche audience with less competition.

Conclusion

The selected keywords "Luxury car rental near me" and "Rolls Royce rental" are strategically chosen based on their strong search volume, low to moderate competition, and high relevance to the business of luxury car rentals. The Instagram ad is designed to attract users looking for premium car rental services by showcasing high-end vehicles, offering a compelling discount, and providing easy booking options. This combination of well-researched keywords and engaging ad design should result in a successful marketing campaign that resonates with the target audience and drives conversions.

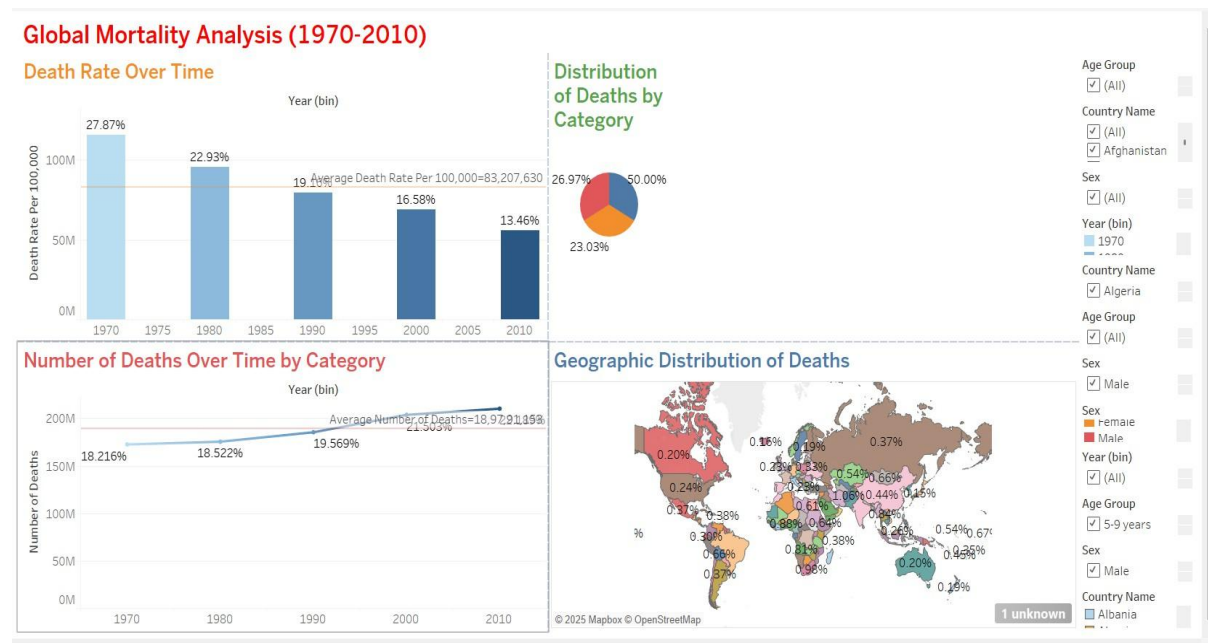
5. Tableau dashboard

1. Objective:

This report presents a Tableau dashboard created using the IHME GBD 2010 Mortality dataset. The dashboard provides insights into mortality trends, including death rates over time, the distribution of deaths by gender, and geographic variations in mortality. The analysis aims to inform public health strategies and resource allocation.

2. Dashboard Overview:

Dashboard Title: Global Mortality Trends Dashboard



Dashboard Layout:

- **Death Rate Over Time:** A bar chart visualizing the death rate per 100,000 people from 1970 to 2010.
- **Distribution of Deaths by Gender:** A pie chart showing the proportion of deaths across genders.
- **Number of Deaths Over Time:** A line chart illustrating the number of deaths over time.
- **Geographic Distribution of Deaths:** A choropleth map visualizing the distribution of deaths across different countries.

Filters:

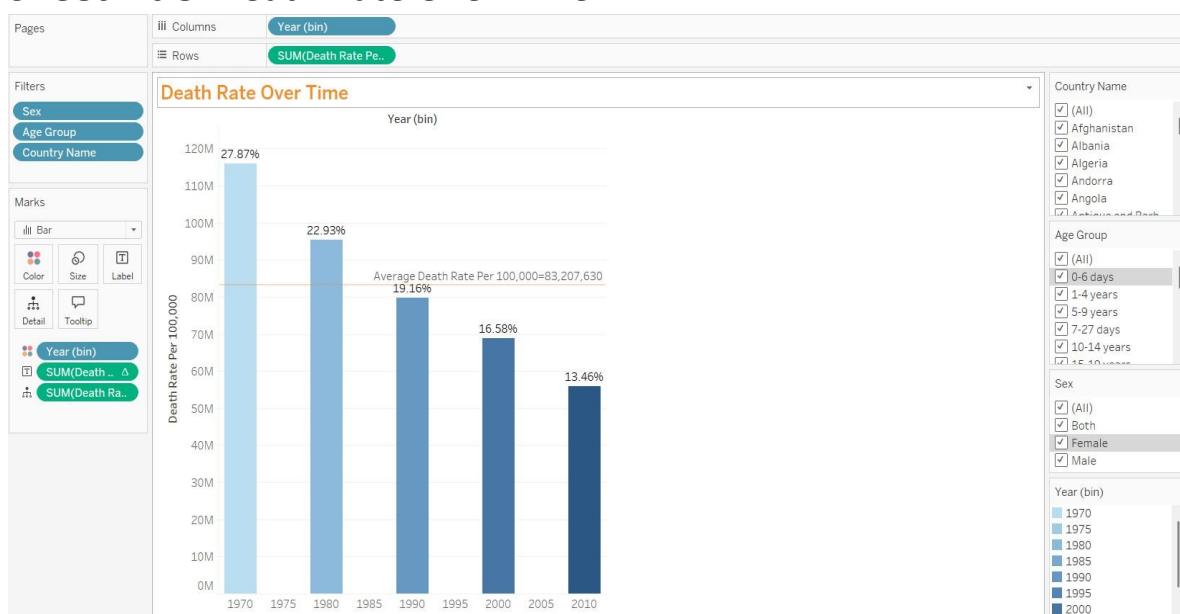
- Year (bin)
- Age Group
- Sex
- Country Name

Overall: This dashboard provides a comprehensive overview of mortality trends and allows users to explore the data from multiple perspectives.

3. Analysis Sections

3.1 Death Rate Over Time

Sheet Title: Death Rate Over Time



Description:

- **Data Source:** IHME GBD 2010 Mortality Dataset.
- **Visualization Type:** Bar Chart.
- **X-axis:** Years (binned) from 1970 to 2010.
- **Y-axis:** Death rate per 100,000 people.
- **Bars:** Each bar represents the death rate for a specific year.
- **Colour:** Bars are coloured using a blue gradient, with darker shades indicating higher death rates.

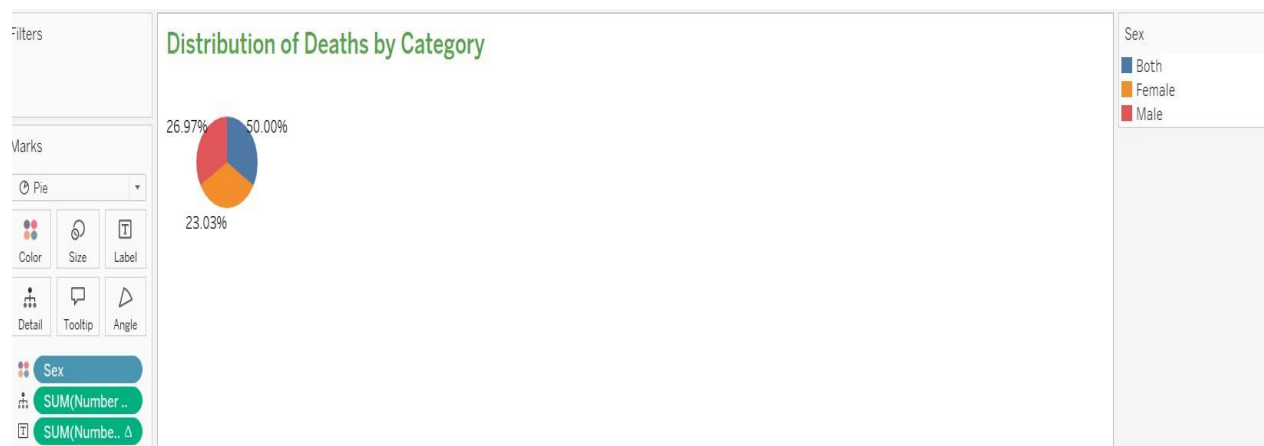
- **Horizontal Line:** Indicates the average death rate across the years.

Key Insights:

- The year **1970** has the highest death rate, standing at **27.87%** of the total death rate per 100,000.
- The year **2010** has the lowest death rate, at **13.46%**.
- The average death rate across the years is indicated by a horizontal line at **83,207,630**.

3.2 Distribution of Deaths by Gender

Sheet Title: Distribution of Deaths by Gender



Description:

- **Data Source:** IHME GBD 2010 Mortality Dataset.
- **Visualization Type:** Pie Chart.
- **Data Representation:** Distribution of deaths across genders.

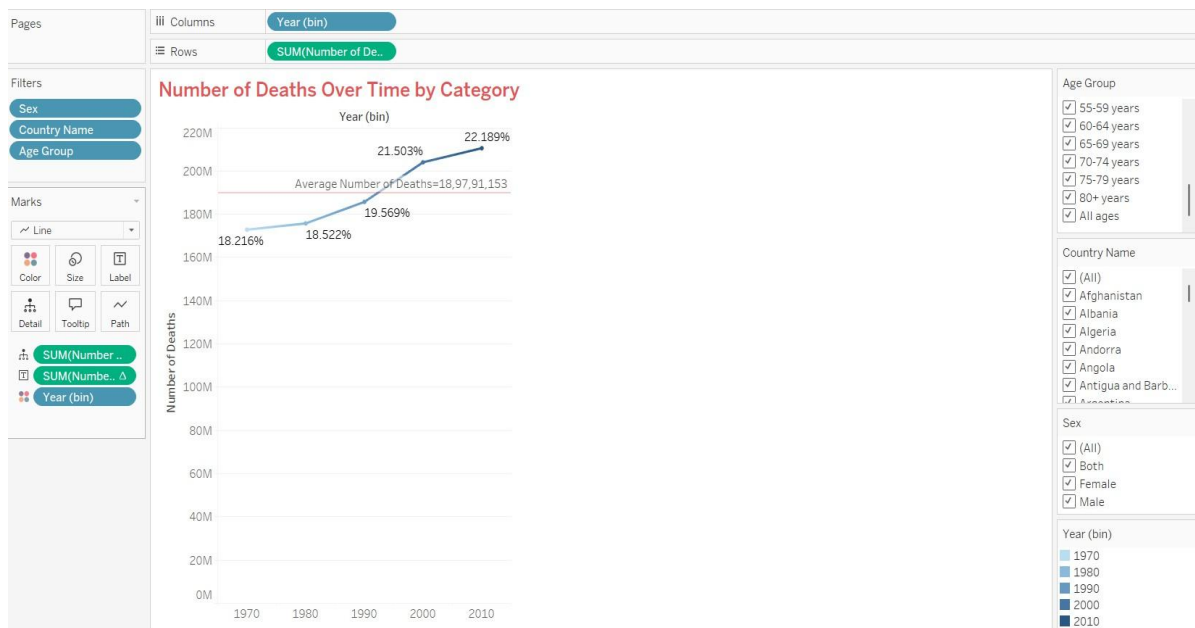
Key Insights:

- The pie chart shows that the share of deaths is distributed as follows:

- **Female: 23.03%** of the total number of deaths.
- **Male: 26.97%** of the total number of deaths.
- **Both: 50%** of the total number of deaths.

3.3 Number of Deaths Over Time

Sheet Title: Number of Deaths Over Time



Description:

- **Data Source:** IHME GBD 2010 Mortality Dataset.
- **Visualization Type:** Line Chart.
- **X-axis:** Years from 1970 to 2010.
- **Y-axis:** Number of deaths.
- **Lines:** Multiple lines representing different categories (e.g., age groups, sex).

Key Insights:

- The year **2010** has the highest percentage of deaths at **21.189%**.

to analyse specific years, age groups, and countries, providing insights into geographic disparities in mortality.

4. Conclusion

The Tableau dashboard created using the IHME GBD 2010 Mortality dataset provides valuable insights into mortality trends over time, the distribution of deaths by gender, and geographic variations. By utilizing various visualizations and filters, users can explore the data in depth, informing public health strategies and resource allocation.